PROJECT CHARTER

Virginia Market Square

Community Vendors Platform — Website Redesign & E-Commerce Build

Course	CIS2987 — Web Development Capstone
Project Name	Virginia Market Square — Community Vendors Platform
Student / Developer	Capstone Student, CIS2987
Instructor	Anup Parajuli
Organization	Virginia Market Square (Non-Profit), Virginia, MN
Project Duration	March 13 – April 26, 2026 (6 Weeks)
Total Hours	52 hours estimated
Version	1.0 — March 2026
Status	✓ **APPROVED**

1. Executive Summary

This project charter formally authorizes the redesign and development of a new website for Virginia Market Square, a community-focused non-profit farmers market located in Virginia, Minnesota. The current Squarespace site limits the organization's ability to showcase vendors, manage content, and scale as the market grows.

The capstone project will deliver a fully custom, database-driven website built with PHP, MySQL, Bootstrap 5, and JavaScript. The advanced scope includes vendor self-service logins, a shopping cart and checkout system, Stripe payment integration, and a normalized 12-table relational database — demonstrating real-world full-stack web development skills.

The project runs for seven weeks (March 16 – May 1, 2026) at an estimated 52 total development hours. At completion, Virginia Market Square will have a live, professional website replacing their current Squarespace platform at a fraction of the long-term hosting cost.

2. Project Background & Justification

2.1 Organization Overview

Virginia Market Square is a non-profit farmers market operating in Virginia, Minnesota, running every Thursday from June through October (2:30–6:00 PM) at 111 South 9th Avenue W. The market supports local farmers, makers, and producers within 50 miles, emphasizing organic, sustainable, and community-first values.

2.2 Current Website Limitations

The existing Squarespace website presents several constraints:

- No vendor self-management — all content updates require admin intervention
- No e-commerce capability — customers cannot browse or purchase products online
- Limited database functionality — no searchable vendor or product directory
- Escalating cost — Squarespace fees increase as features are added
- No admin control panel — content cannot be managed without Squarespace access

2.3 Business Justification

Building a custom website delivers lasting value to Virginia Market Square:

- Vendors can self-manage their profiles and product listings
- Customers can browse and purchase products online via a shopping cart
- Hosting costs reduce to approximately $100–200/year vs. Squarespace subscriptions
- The organization owns the codebase and can extend it without vendor lock-in
- A scalable foundation supports future features such as mobile apps and newsletters

3. Project Objectives

This project has three categories of objectives:

Category	Priority	Objective
Technical	**Must**	Deliver a fully responsive website using Bootstrap 5 that works across all devices
Technical	**Must**	Implement a normalized 12-table MySQL database with foreign keys and proper 3NF design
Technical	**Must**	Build authentication for three user types: Admin, Vendor, and Customer
Technical	**Must**	Create a functional shopping cart and multi-step checkout flow
Technical	**Must**	Integrate Stripe payment processing (test mode)
Technical	**Should**	Build a vendor self-service dashboard with sales analytics
Business	**Must**	Replace the existing Squarespace site with a fully owned, custom platform
Business	**Must**	Enable admin to manage vendors, events, and contact submissions
Business	**Should**	Provide vendors with tools to manage their own products and view orders
Educational	**Must**	Demonstrate proficiency in PHP, MySQL, JavaScript, and full-stack development
Educational	**Must**	Apply database normalization and complex SQL JOINs in a real-world context
Educational	**Must**	Complete a portfolio-quality project suitable for professional presentation

4. Project Scope

4.1 In Scope

- Homepage with featured vendors and upcoming market events
- Vendor directory with search and category filtering
- Individual vendor detail and vendor storefront pages
- Product catalog with category browsing and product detail pages
- Database-backed shopping cart (persists across sessions)
- Multi-step checkout: Shipping → Review → Payment → Confirmation
- Stripe payment integration (test mode)
- Customer account registration, login, and order history
- Vendor self-service login, dashboard, product management, and order tracking
- Admin login with management of vendors, events, applications, and contacts
- Vendor application form with admin approval workflow
- Events calendar displaying market dates and special events
- Contact form with vendor-specific inquiry routing
- About, Mission, and informational static pages
- Fully responsive Bootstrap 5 design for mobile, tablet, and desktop
- 12-table normalized MySQL database with indexes and foreign keys
- CSRF protection, password hashing (bcrypt), SQL injection prevention
- Deployment to live production hosting with SSL certificate
- Complete documentation including README, admin manual, and database schema

4.2 Out of Scope

- Native mobile application (iOS / Android)
- Real Stripe payments — test mode only for this capstone
- Automated email notification system (basic PHP mail only)
- Product reviews and ratings system
- Inventory alert / low-stock notification system
- Customer wishlist feature
- Advanced analytics dashboard with charting beyond vendor metrics
- Integration with external platforms (social media, Google Analytics)
- Multi-language / internationalization support

5. Project Deliverables

#	Deliverable	Phase	Target Date
1	Project charter, sitemap, wireframes (22+ pages), visual mockups	Phase 1 — Planning	**Mar 13, 2026**
2	Advanced ER diagram (12-table schema, 3NF) and database design document	Phase 1 — Planning	**Mar 13, 2026**

3	Complete database.sql with all 12 tables, indexes, and test data	Phase 2 — Database	**Mar 20, 2026**
4	Working authentication for Admin, Vendor, and Customer user types	Phase 3 — Auth	**Mar 22, 2026**
5	Public product browsing pages with complex SQL JOINs	Phase 4 — Backend	**Apr 1, 2026**
6	Vendor storefront, category pages, and search functionality	Phase 4 — Backend	**Apr 1, 2026**
7	Shopping cart (database-backed) with add, update, and remove	Phase 4 — Backend	**Apr 5, 2026**
8	5-step checkout flow and order creation from cart	Phase 4 — Backend	**Apr 5, 2026**
9	Fully responsive Bootstrap design across all 25+ pages	Phase 5 — Payment	**Apr 12, 2026**
10	Stripe payment integration (test mode) with error handling	Phase 6 — Frontend	**Apr 14, 2026**
11	Vendor dashboard with sales analytics and product management	Phase 4/6 — Combined	**Apr 23, 2026**
12	JavaScript cart interactivity, form validation, and AJAX updates	Phase 7 — JS	**Apr 23, 2026**
13	Populated database with 8–10 vendors and 40–50 products	Phase 8 — Content	**Apr 24, 2026**
14	Live deployed website with SSL on production hosting	Phase 9 — Deploy	**April 26, 2026**
15	README, admin manual, vendor manual, and project report	Phase 9 — Deploy	**April 26, 2026**

6. Project Timeline

Week	Dates	Hours	Focus & Key Activities	Phases
1	**Mar 13-16**	**6 hrs**	Planning, wireframes (22+ pages), ER diagram, database schema design, branding guidelines	*Phases 1–2*
2	**Mar 16-23**	**7 hrs**	Write database.sql (12 tables), populate test data, build all 3 authentication systems	*Phases 2–3*
3	**Mar 23-30**	**8 hrs**	Product browsing with JOINs, vendor storefront, category pages, search functionality	*Phase 4 (start)*
4	**Apr 30-6**	**8 hrs**	Shopping cart (add/update/remove), multi-step checkout, order creation from cart	*Phase 4 (cont.)*

5	Apr 6-13	8 hrs	Bootstrap responsive design for all 25+ pages, mobile testing	*Phase 6*
6	Apr 13-20	8 hrs	Stripe payment integration, vendor dashboard with analytics, JS/AJAX interactivity	*Phases 5, 7*
7	Apr 20-27	7 hrs	Sample content, comprehensive testing, bug fixes, deployment, documentation	*Phases 8–9*
∑	**Mar 16 – May 1, 2026**	**52 hrs**	**Total Project Hours**	**All 9 Phases**

7. Technology Stack

Technology	Version / Tool	Purpose
HTML5	*Semantic HTML5*	Document structure, accessible markup throughout all pages
CSS3	*Custom + Bootstrap 5*	Styling, responsive grid, component library, branding overrides
Bootstrap	*Bootstrap 5.3 (CDN)*	Responsive layout, navbar, cards, forms, modal components
JavaScript	*Vanilla ES6+*	Form validation, AJAX cart updates, interactive UI elements
PHP	*PHP 8.x*	Server-side logic, routing, session management, form handling
MySQL	*MySQL 8 / MariaDB*	Relational database — 12-table normalized schema (3NF)
Stripe	*Stripe PHP SDK*	Payment processing (test mode) — PCI-compliant tokenization
Apache	*XAMPP (local) / cPanel*	Local development server and production web hosting
Git	*Git + GitHub*	Version control, code backup, project history
Composer	*Composer 2*	PHP dependency management (Stripe SDK installation)

8. Risks & Mitigation

Risk	Likelihood	Impact	Mitigation Strategy
Scope creep — adding features mid-project	Medium	High	Adhere strictly to the priority tiers defined in scope; defer Nice-to-Have features to Phase 2

Stripe integration more complex than estimated	Medium	Medium	Allocate full Week 6 for payment; fall back to PayPal Checkout if Stripe SDK issues arise
Database schema requires major redesign mid-project	Low	High	Finalize and validate the 12-table ERD in Week 1 before writing any PHP pages
Running behind schedule by Week 4	Medium	High	Launch with Priority 1 (35 hrs) scope — skip Stripe and use a placeholder payment page
Hosting/deployment issues at launch	Low	Medium	Test on a staging subdomain in Week 6; deploy at least 48 hours before May 1 deadline
Security vulnerabilities (SQL injection, XSS)	Low	High	Use prepared statements throughout; apply htmlspecialchars() on all output; CSRF tokens on forms
Bootstrap responsiveness issues on mobile	Medium	Low	Test on real devices weekly from Week 5; use browser DevTools mobile emulator daily

9. Success Criteria

The project will be considered successfully complete when ALL of the following criteria are met by May 1, 2026:

	Success Criterion	Category
✓	Website is live at a real domain and publicly accessible	*Deployment*
✓	All three user types can log in (Customer, Vendor, Admin)	*Authentication*
✓	Product browsing works with SQL JOINs pulling vendor and category data	*Database*
✓	Shopping cart stores items in the database and persists across sessions	*E-Commerce*
✓	5-step checkout process completes and creates orders in the database	*E-Commerce*
✓	Stripe payment processes test transactions successfully	*Payment*
✓	Orders are created correctly in the orders and order_items tables	*Database*
✓	Vendor dashboard displays accurate sales metrics via SQL aggregation	*Analytics*
✓	Vendors can add, edit, and delete their own products only	*Vendor Portal*
✓	Website is fully responsive on mobile (375px), tablet (768px), and desktop	*Frontend*
✓	All 25+ pages are styled consistently using Bootstrap 5	*Frontend*
✓	Database uses proper normalization with foreign keys and no redundant data	*Database*
✓	Passwords are hashed with bcrypt — no plain-text passwords stored	*Security*

✓	All forms are protected with CSRF tokens	*Security*
✓	All SQL queries use prepared statements to prevent injection	*Security*
✓	Complete documentation delivered: README, admin manual, vendor manual, DB schema	*Documentation*
✓	Test data populated: 8–10 vendors, 40–50 products, sample orders and events	*Content*

10. Charter Approval & Sign-Off

By signing below, the parties acknowledge they have reviewed this project charter and authorize the project to proceed as described.

Role	Name	Date
Student / Developer	_____ Signature: _____	_____
Course Instructor	_____ Signature: _____	_____
Organization Contact	_____ Signature: _____	_____

Document Version: 1.0 | Created: March 2026 | Last Updated: March 2026 | Status: Approved